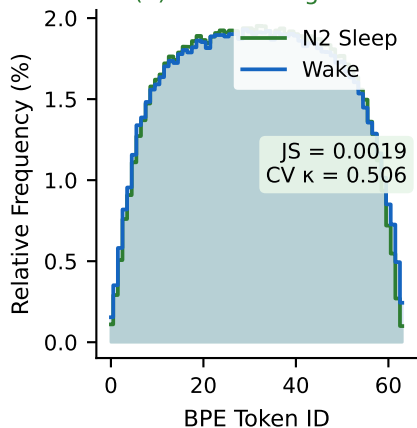
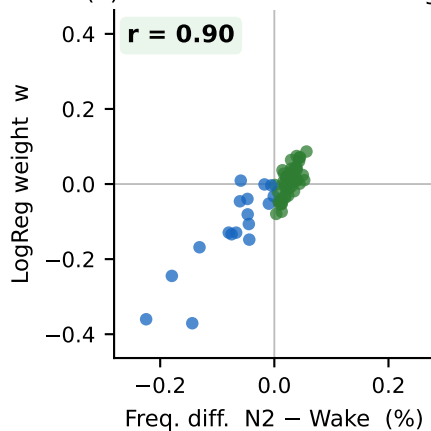


Sleep-EDF (amplitude-coded)

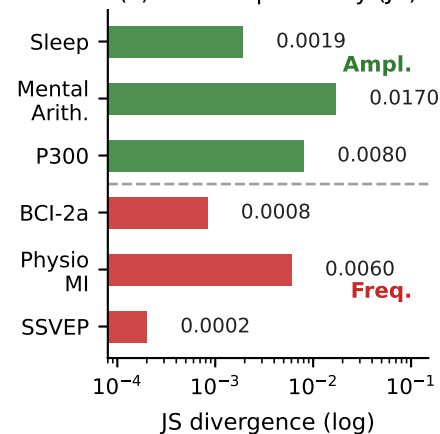
(a) Class histograms



(b) Token enrichment vs weight

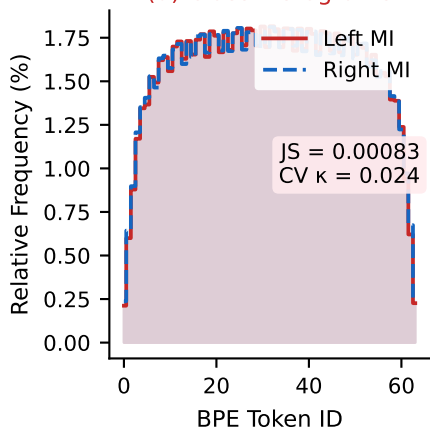


(c) Class separability (JS)

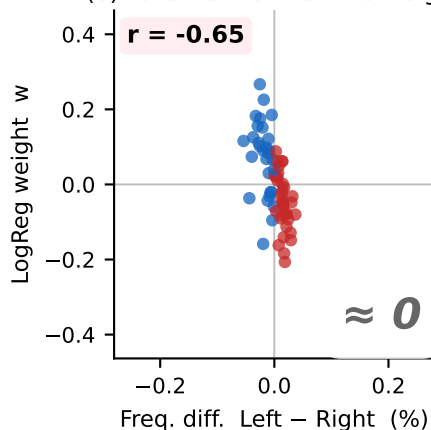


BCI-IV-2a (frequency-coded)

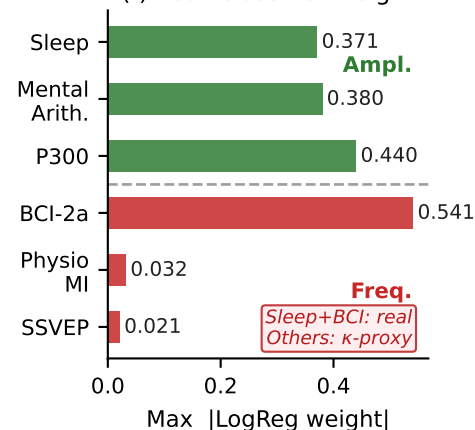
(d) Class histograms



(e) Token enrichment vs weight



(f) Peak classifier weight



(a)-(d): real BPE histograms (real/real; $V=1024$, $B=64$, adaptive, ≤ 300 epochs/class). (c)/(f): Sleep+BCI real; others from κ -proxy.